

# Shenyu Qin

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## Education

### Shanghai Jiao Tong University

Sept. 2022 – Present

*Undergraduate in Computer Science (John Hopcroft Honors Class)*

- GPA: 3.78/4.3 (89.03/100)
- Relevant Coursework: Mathematical Analysis II (93), Combinatorial Mathematics (91), Probability Theory (91), Computational Complexity (98), Online Algorithms (95), Statistical Learning Theory (97)

## Research Experience

### Pandora's Box Problem with Deadlines

John Hopcroft Center, SJTU

*Advised by Prof. Yuhao Zhang*

Jan. 2026 – Present

- Ongoing

### Existence of $\alpha$ -winning and $\alpha$ -dominating committees in an Impartial Culture

DIMACS, Rutgers University

Aug. 2025 – Dec. 2025

*Advised by Prof. Kangning Wang and Prof. Lirong Xia*

- Studied the existence of  $\alpha$ -winning and  $\alpha$ -dominating committees of size  $k$  in an impartial culture when the number of candidates  $m$  is much larger than the number of voters  $n$
- Established two sharp thresholds for  $\alpha$ : an  $\alpha$ -winning committee exists w.h.p. iff  $\alpha < \frac{k-1}{k}$ , and an  $\alpha$ -dominating committee exists w.h.p. iff  $\alpha < \frac{k-1}{2k}$ , when  $m$  is sufficiently larger than  $n$  and  $n = \omega(1)$ .
- Wrote a manuscript based on the results

### Online Load and Graph Balancing for Random Order Inputs

John Hopcroft Center, SJTU

*Advised by Prof. Yuhao Zhang*

Jul. 2024 – Mar. 2025

- Studied the online load balancing problem for random order inputs, aiming to close the gap between the  $\Omega(\sqrt{\log m})$  lower bound and the  $O(\log m / \log \log m)$  upper bound by designing an  $O(\sqrt{\log m})$ -competitive algorithm, where  $m$  is the number of machines.
- Investigated the online graph balancing problem, which is a special case of the original problem, trying to analyze the competitive ratio of a newly-designed algorithm in this case
- Participated in TCS Reading Group every week

## Preprints

### Winning in the Limit: Average-Case Committee Selection with Many Candidates

[arXiv](#)

*with Yifan Lin, Kangning Wang and Lirong Xia*

## Teaching Experience

Design and Analysis of Algorithms (CS3950), Teaching Assistant, SJTU

Spring 2026

## Honors and Awards

Zhiyuan Honors Scholarship (Top 10%), SJTU

2022, 2023, 2024, 2025

## Skills

**Programming** C++, C, Python, Rust

**Tools:** L<sup>A</sup>T<sub>E</sub>X, MATLAB, Coq

**Languages:** Mandarin (Native), English (TOEFL: 104)